

CMS $137.6 \text{ fb}^{-1} (13 \text{ TeV})$

2HDM+S Type-IV

 $m_H = 125 \text{ GeV}$

95% CL upper limits

- Expected exclusion
- Observed exclusion
- Expected exclusion
- Observed exclusion
- Expected exclusion
- Observed exclusion
- Expected exclusion
- Observed exclusion

 $B(H \rightarrow aa) = 1.0$ $B(H \rightarrow aa) = 0.16$ $\frac{\sigma_H B(H \rightarrow aa)}{\sigma_{SM}}$ 10^{-1} 10^{-2} 10^{-3} $\tan \beta$ 